DevOps & AWS Cloud Technical Handbook

1. AMAZON SQS (SIMPLE QUEUE SERVICE)

Amazon SQS ek fully managed message queuing service hai jo microservices aur distributed applications ko aapas mein decouple aur scale karne ke kaam aati hai. Jab aapki application par sudden requests ka traffic bohot zyada badh jaata hai, toh SQS un requests ko ek buffer queue mein safe rakh leta hai taaki backend processes bina crash hue apni processing capacity ke hisab se dheere-dheere unhe consume kar sakein.

Core Components Aur Technical Limits:

- **Batch Processing Architecture:** Isme standard runtime configurations ke mutabik aap ek single API invoke call (SendMessage, ReceiveMessage, DeleteMessage) ke andar maximum 10 messages tak ka data chunk ek sath processes kar sakte hain.
- **Throughput Parameter:** Standard queues mein messages per second ki koi upper limit nahi hoti, yeh instantly scale hoti hain. FIFO queues default roop se 300 messages/sec support karti hain aur batching ke sath high configurations par ja sakti hain.
- **Payload Size Optimization:** Ek continuous single packet structure ka physical raw size data limit strict 256 KB tak bounded hota hai. Badi binary objects ke liye data clusters Amazon S3 arrays par pass out kiye jaate hain aur actual queue references ke pass bas marker index store rehta hai.
- **Queue ke Types:** Standard Queues messages ko kam se kam ek baar deliver karne ki guarantee deti hain par unka sequence change ho sakta hai. FIFO (First-In-First-Out) Queues strict ordering aur exactly-once processing ensure karti hain jisse data duplicate nahi hota.

2. AMAZON SNS (SIMPLE NOTIFICATION SERVICE)

Amazon SNS ek asynchronous Pub/Sub (Publish/Subscribe) notification service hai. Iska kaam kisi bhi message ko publish hote hi instantly fan-out model ke zariye sabhi subscribers tak push notification ke roop mein pahunchana hai.

Core Components Aur Features:

- **Topics Configuration:** Isme topics main entry points bante hain jahan server producers applications apney log status aur data alerts publish karte hain.
- **Fan-Out Broadcaster Engine:** SNS Topic par jaise hi koi event hit karta hai, yeh ek sath multiple destinations jaise ki Amazon SQS queues, AWS Lambda functions, HTTP/HTTPS webhooks, SMS gateways, aur Emails par data broadcast kar deta hai.

3. AMAZON SES (SIMPLE EMAIL SERVICE)

Amazon SES ek highly affordable aur reliable cloud-based email sending service hai. Yeh application developers ke liye infrastructure handle karti hai taaki direct SDK ya SMTP integrations ke zariye automatic aur scalable emails bheje ja sakein.

Core Components Aur Use Cases:

- **SMTP Endpoints & Verified Identities:** Yeh secure user authentication, custom domains verification arrays, aur clean IP routing mechanisms manage karta hai taaki mail server boundaries handle ho sakein.
- **Mailing Workflows:** Iska upayog transactional emails (jaise password reset verification links, immediate account alerts) aur large marketing distribution campaigns ke liye kiya jata hai.

4. VELERO SUBSYSTEM (KUBERNETES BACKUP & DISASTER RECOVERY)

Velero ek open-source cloud utility hai jisse Kubernetes cluster ke poore infrastructure state, configuration objects, aur persistent application volumes ka backup liya jata hai aur disaster recovery execute ki jaati hai.

Core Components:

- **Velero Server Daemon:** Yeh Kubernetes cluster ke andar ek manager controller ki tarah chalta hai jo active CRDs aur configurations ke lifecycle ko manage karta hai.
- **Velero CLI Tool:** Local terminal control package jisse immediate cluster backup runs ya manual verification tasks execute kiye jaate hain.
- **Storage Provider Plugins:** Modular code integration bundles jo data snapshots ko directly external secure cloud storage layers par ship karte hain.

5. MINIO OBJECT STORAGE ENGINE

MinIO ek high-performance, open-source object storage server framework hai jo framework pipelines ke liye Amazon S3 ke standard cloud APIs se 100% compatible hota hai.

Core Components:

- **MinIO Core Binary Services:** Background data nodes par continuous object transactions maintain karne wala high speed server array storage application node space hai.
- **MinIO Terminal Client CLI:** Custom commands executions scripts configurations run karne aur tracking check metrics filters target profiles verify karne ka control package dashboard tool path hai.
- **Distributed Clustered Mode:** High storage resilience setups achieve karne ke liye localized multi-nodes drives arrays systems configurations tracking networks use karta hai.

6. AMAZON ECR (ELASTIC CONTAINER REGISTRY)

Amazon ECR AWS ki ek fully managed Docker private container registry service hai jahan developers apni container images ko completely secure, isolated aur organized tarike se store kar sakte hain.

Core Components Aur Properties:

- **Image Registry Repositories:** Projects layout structure base path jahan systems engineering images properly categorized formats mein securely preserve hoti hain.
- **Automated Scanning Engine:** Yeh image upload hote hi native automated vulnerability scanning triggers chalta hai jo CVE code check karke software vulnerabilities alert karta hai.
- **Lifecycle Auto Policies:** Automate cleaning execution modules jo historical ya unused image tags ko auto-delete karke memory storage cost optimize karti hain.

7. AMAZON ECS (ELASTIC CONTAINER SERVICE)

Amazon ECS AWS ka ek built-in native container orchestrator platform hai jo Docker containers ko scale karne aur cloud instances par distribute karne ke kaam aata hai.

Core Components Aur Tasks Layout:

- **ECS Cluster Setup:** Environment instances groups platforms layers handle karne ka central primary logical configuration boundary space networking map block layout zone.
- **Task Definition Schemas:** Strict structured programmatic JSON syntax layout templates script parameters files, jisme computing allocations parameters, image build repositories source paths maps, port mapping links parameters, environment parameters keys configuration settings define kiye jaate hain.
- **Active Running Tasks:** Structural operational system nodes instances runtime applications configurations components execution objects parameters (Active Running Containers blocks).
- **Orchestrator Execution Services:** Managed replica sets maintenance control layers framework systems jo active running containers health status parameters, automatic container auto scaling targets properties mappings, cloud network routes connectivity options load balancers mappings completely manage aur tracking checks automate karti hain.

8. AMAZON EKS (ELASTIC KUBERNETES SERVICE)

Amazon EKS ek fully managed Kubernetes distribution environment hai jo control plane high availability, etcd storage syncing aur node components ke standard maintenance burdens ko completely automate aur abstract kar deta hai.

Core Components:

- **EKS Core Managed Control Plane:** Kubernetes API server management systems, high availability etcd nodes synchronization routines, master systems patches deployment, aur configurations logs handling pipelines architecture configurations.
- **EKS Computing Worker Nodes Groups:** Operating machine layouts computer target systems clusters servers configurations groups compute arrays maps jahan clusters applications pod configurations containers blocks live production execution models processes control paths use karte hain (Managed EC2 pools servers or serverless profiles ke zariye AWS Fargate clusters connect kar sakte hain).

9. KARPENTER (ADVANCED POD SCALE ENGINE)

Karpenter ek high-speed, flexible Kubernetes cluster autoscaler framework hai jo conventional compute configurations ke complex steps aur delays ko directly bypass karta hai.

Core Components Aur Mechanisms:

- **Direct Cluster API Monitor:** Intelligent cluster configurations state changes tracker loop subsystem parameters jo Kubernetes cluster pending setups applications scheduling patterns metrics evaluate karta hai.
- **Dynamic Provisions Controller:** Custom definitions criteria templates files properties codes configuration constraints frameworks labels properties targets (`Provisioners` / `NodeClasses`), jo cluster requirements assess karke standard restrictive slow virtual machines templates EC2 Auto Scaling Groups configurations parameters steps delays completely bypass karke direct live clouds architectures layers framework par exact raw instances machine scale resources groups configure launch data updates seconds mein pass out kardeta hai.

10. PODMAN CONTAINER MANAGER

Podman ek open-source container engine hai jo rootless aur completely daemonless setups par local microservice containers run karne ke liye ek advanced Docker alternative banta hai.

Core Tooling Components:

- **Podman User Space CLI:** Front-end execution control utility application setup interface package commands tool path terminal inputs tools.
- **Conmon Process Engine:** Short lifecycle process tracking system framework process execution loops variables monitors nodes tools indicators metrics software blocks components.
- **OCI Compliant Core Runtime:** Low layer open container specification translation platforms system executors applications blocks execution paths models tools (Specialized standard modules systems codes frameworks engines jaise `runc` ya `crun` engines paths configurations layers maps).

11. GITLAB CI/CD INFRASTRUCTURE

GitLab CI/CD ek automatic execution infrastructure framework hai jo repository platform ke andar built-in pipelines run karne ke liye pure lifecycle automation rules trigger karta hai.

Core Configuration Components:

- `.gitlab-ci.yml` **structural config:** Project core configuration codebase manifest scripts code data paths, jahan pipelines processing states automation validation criteria workflows steps sequentially defined locked rehte hain.
- **Pipeline Logical Stages:** Phase divisions logical borders parameters (jaise compile task phases `build` options logic setups validations frameworks tests script iterations `test` deployments triggers rules code environments setups options paths loops options steps `deploy`).
- **Pipeline Structural Jobs:** Single isolated atom actions codes bash shell script parameters instruction sequences execution blocks tools inside specific stage components.
- **GitLab Pipeline Runner agents:** Continuous computation processing background systems platforms daemons execution systems nodes packages models tools runner modules loops agents variables.

12. JENKINS AUTOMATION SERVER

Jenkins ek extensively modular open-source continuous integration automation deployment server hai jo scale distribution architectures ke liye distributed Master-Agent topology patterns utilize karta hai.

Core Subsystems Aur Components:

- **Jenkins Primary Central Master engine:** Logic scheduler plugins managers parameters filters parameters web dashboards view maps controls framework monitoring tracking workflows systems controller engine configurations.
- **Jenkins Remote Agents Systems:** Isolated compute remote machines computing workers execution targets spaces platforms targets node systems codes processing units nodes environments execution agents actual heavy commands execution lines maps parameters configurations parameters.
- **Declarative Jenkinsfile Model:** Programmatic code track configurations patterns systems pipelines scripts format framework configuration layout definitions files code base options tracks inside project repository codebase (Declarative Groovy systems syntax constructs options formats fields tracks).
- **Webhooks Automated Triggers rules:** Direct immediate endpoints metrics triggers alerts hooks (Git code changes updates hits land continuous instant triggers automation paths signals pipelines parameters alerts checks pipelines build tracks start configurations loops parameters settings updates).
- **Secure Credentials Management Panel:** Highly secure encrypted memory blocks configuration framework maps vault profiles, jahan password encryption tokens, authorization paths API access keys, secure connection profiles certificates profiles completely secure isolated layers me preserve rehte hain, aur execution parameters software pipelines processes output outputs tracking screens logs outputs se absolutely cleanly mask hide filters systems layers properties apply karte hain.

13. AWS CODEPIPELINE WORKFLOW

AWS CodePipeline ek fully managed serverless continuous delivery engine execution controller hai jo complex code deployments ke release checkpoints ko track aur map karne ke liye built-in automation layers apply karta hai.

Core Functional Components:

- **Pipeline Release Engine:** Core execution state machine parameters definitions flow paths, jahan source retrieval code versions change transitions parameters update operations cleanly track mapping systems controllers enforce rules settings.
- **Pipeline Stages Execution Segments:** Isolated block tracking levels variables (Source updates checks, Build compilation steps code, Test arrays validation paths updates, Deploy production integrations lines maps profiles models platforms).
- **Actions Parameters Handlers:** Specific task blocks commands definitions execution lines instructions maps variables (jaise pulling repositories packages, run dynamic builds compilation rules code via `buildspec.yml` files settings configurations, code ship artifacts targets maps profiles updates models).

14. TERRAFORM INFRASTRUCTURE AS CODE

Terraform HashiCorp ka ek multi-cloud declarative Infrastructure as Code declarative language execution suite hai jo cloud properties ko HCL syntax formats mein save karke state configurations manage karta hai.

Core Components:

- **Terraform Core Engine:** Logic state compilation dependency tree maps builder code parameters variables reconciliations engine solution suite framework applications modules logs tracing data management profiles tracks options fields properties targets parameters.
- **Cloud Platform Providers Plugins:** Target API structural maps plugins translation models specifications dependencies packages profiles engines layers (Specialized target engines maps providers structures codes setups frameworks lines modules updates models profiles targets variables maps jaise AWS systems maps, Azure cloud systems filters profiles models targets parameters, GCP modules maps).
- **Cloud Resources Structures Objects Declarations:** Declarative resource blocks parameter metrics specifications code layout profiles schemas blocks maps variables settings systems definitions components lines paths configuration.
- **State Tracking Layout Databases (`.tfstate` configuration):** Continuous active tracking file data index logs records layout parameters map, jahan cloud real world entities details allocations configurations values variables parameters maps properly current code configurations patterns maps synchronize lock profiles records data formats tracking file.

```
# Initialize infrastructure directories and configure backend provider
modules
terraform init

# Check code files structural updates against actual resource
configurations
terraform plan

# Deploy infrastructure adjustments directly to live production spaces
terraform apply -auto-approve

# Cleanly terminate and destroy all virtual cloud systems handled here
terraform destroy -auto-approve
```

15. ANSIBLE AUTOMATED CONFIGURATION SYSTEM

Ansible ek open-source, completely agentless configuration system controller engine hai jo remote target virtual servers ko secure SSH or WinRM lines par dynamic configuration definitions ke sath manage aur bootstrap karta hai.

Core Components Aur Parameters:

- **Ansible Primary Central Control Node System Machine:** Main software deployment installation workspace path platform shell host jahan execution engine packages variables lines scripts models platforms profiles run parameters definitions controls command lines tools are managed.
- **Target Remote Managed Nodes Endpoints Systems:** Systems infrastructure processing units endpoints hosts instances groups machines network profiles structures profiles targets parameters configurations variables lines parameters managed servers.
- **Inventory Configuration Mapping Listings Dataset Files (`inventory.ini`):** Core documentation records structures listings file profiles jahan server grouping structures variables codes network access parameter strings target hosts metadata configurations profiles records fields settings data structures documents configurations.
- **Independent Execution Modules Commands Files:** Automation actions units components code snippets specifications libraries files, jo configuration operations remote machine levels target spaces check tasks modify adjustments systems metrics criteria parameters map tools properties filters layers.
- **Structured Automation Workflows Code Paths Files (`Playbooks`):** Clean code structures patterns design blueprints configuration manifests workflows maps logic guidelines script files written in human clean code structures syntax configurations formatting YAML parameters configurations files records data layouts tracks properties lines options settings field.
- **Conditional Triggers Handlers Mechanisms Parameters Filters Targets:** Specialized event handlers logic criteria conditional automation steps execution signals codes modules loops, jahan change modifications events indicators outputs catches traces land (`changed` states markers triggers), immediately precise operational secondary tasks steps definitions targets alerts signals (jaise restarting custom systems web daemons servers instances

profiles, flushing systems buffers indexing paths configuration updates configurations parameters status parameters indicators code items check metrics settings updates).

```
# Instantly test baseline ping visibility parameters over target tracking
scopes
ansible all -m ping -i inventory.ini

# Trigger structured automation blueprints written inside YAML execution
maps
ansible-playbook -i inventory.ini main_site_deployment.yml

# Fetch modular configuration elements and roles from community tracking
zones
ansible-galaxy collection install community.aws
```

16. GIT & GITHUB VERSION CONTROLS

Git ek locally distributed tracking version repository application hai jo software operations code updates aur code changes ki full history manage karti hai. GitHub isi tracking setup ko online support dene aur team pull requests manage karne ka central cloud web platform hai.

Core Workflow Operations Components:

- **Local Tracking Repository Directory Index:** Projects directory tracking folder database structures index storage framework patterns files, jahan structural adjustments history profiles logs updates versions changes records files securely preserve metadata configurations information paths systems configuration files records templates configurations formats fields layouts profiles.
- **Tracking Snapshots Permanent Indexes (`Commits`):** Permanent tracking database records identifiers strings logs, jahan historical data checkpoints structural codes states profiles versions changes updates snapshots records logs values descriptions parameters.
- **Isolated Lines Development Boundaries (`Branches`):** Parallel tracking workflow configurations options tracks paths fields, jahan software modifications feature runs development loops steps lines safely live environments tracks isolate completely clean root baseline lines tracks properties options fields settings.

```
# Initialize an empty Git version storage zone within the current path
git init

# Stage dynamic code modifications for tracking integration steps
git add .

# Save staged modifications permanently to tracking files with context
logs
git commit -m "feat: complete microservice decoupled queue patterns"

# Send active tracking checkpoints out to a remote cloud repository path
git push origin main
```

17. PROMETHEUS & GRAFANA SYSTEM MONITORING

Prometheus ek dynamic time-series metrics monitoring aur automated alerting toolkit engine hai jo continuous targets se statistics data pull model frameworks ke zariye collect karta hai. Grafana ise visual look dene ke liye metrics panels dashboards par live chart systems ke zariye monitor karne ke kaam aata hai.

Core Metrics Scraping Components:

- **Prometheus Scraping Core Engine Server System:** Metric pull systems parameters scraping execution framework, jahan data streams time counts targets snapshots data periodic intervals par HTTP endpoints pull model frameworks through memory data tables store metadata profiles maps logs queries.
- **Target Metrics Exposures Translators (`Exporters`):** System translation agents tools bundles software modules configurations specifications, jo standard system states metrics parameters configurations parameters third party systems profiles indicators (jaise Linux OS resources parameters, MySQL query loops counts metrics, Nginx connection strings analytics) Prometheus query formatting styles text outputs lines layout transform clear metrics read indicators paths configuration parameters data metrics settings.
- **Advanced Metrics Analytic Validation Syntax Language (`PromQL`):** Metric querying structures math vectors filtering expressions functions code syntax layout language metrics queries validation blocks profiles tracking rules data options fields fields properties fields criteria.
- **Grafana Analytical Multi Queries Dashboards Dashboard Layouts:** Multi panels visual screens canvas dashboards tracking systems data visualizer plots line charts graphs mappings tools panels widgets maps, jahan dynamic metrics query scripts outputs metrics parameters live graphs displays track monitors data visual look dashboards alert metrics parameters thresholds settings updates configurations criteria variables parameters.

18. TRIVY VULNERABILITY SECURITY SCANNER

Trivy ek ultra-fast enterprise security scanner platform application tool suite hai jo dynamic containers deployment se pehle active container images, file directory dependencies, aur Iaak definition files ke andar security flaws, CVE issues, aur secrets check karta hai.

Core Components:

- **Scanner Core Engine Binary Execution Paths:** Vulnerability matching checking logic algorithms engines solutions codebase paths profiles tracks fields tools, jo system image layout blueprints targets layers dependencies libraries configuration definitions files compare code matching CVE vulnerability global index data tracks files information security flaws checks alerts.
- **CI/CD Pipeline Automation Wrappers Deployment Loops:** Automation steps actions definitions templates parameters script blocks codes configuration profiles hooks tasks layers, jahan production pipeline builds deployment sequences levels configurations triggers run image build output layers checks pipelines security scans gates blocks targets block control indicators parameters trace metrics code values check.

```
# Scan a targeted docker container infrastructure image tracking block
trivy image node:alpine

# Scan local repository config directories for hidden misconfigurations
trivy config ./infrastructure-state/
```

19. NGINX WEB SERVER ARCHITECTURE

Nginx ek asynchronous, high-performance event-driven reverse proxy, structural web traffic load balancer, aur static files caching web infrastructure server system daemon engine hai.

Core Components:

- **Nginx Central Master Daemon Service Process:** Master tracking daemon control system process parameters definitions flow targets, jahan system settings configuration mappings read, configurations check validation loops steps tracks, child actions processes allocations signals handling layers completely manage control options paths.
- **Asynchronous Non-Blocking Connection Handlers (`Worker Processes`):** High speed isolated execution context processes models compute nodes operations loops tracking, jahan thousands incoming traffic connections lines multi requests flows network requests triggers run event loops resource tracking metrics elements footprints low consumption blocks parallel memory usage speed instant performance models.

20. REDIS IN-MEMORY CACHE

Redis ek extreme performance open-source in-memory network data structure storage array system engine hai jo database layouts, caching layers, aur message brokers ke roop mein operate karta hai.

Core Components:

- **In-Memory Computation Storage Data Grid Engine Block:** Raw access context data storage computing memory areas systems modules variables grids, jahan operations tracking reading writing actions speed high performance layers direct RAM blocks allocations fetch execution parameters optimization tasks run instantly pipelines micro-seconds response times.
- **Advanced Data Structures Models Configuration Types:** Structural internal options data tracking layout definitions variable types items configurations profiles maps (Specialized core entities targets fields parameters specifications variables patterns settings systems configurations tracks jaise standard basic values data `Strings`, map pairs values keys maps collections sets `Hashes`, sequential line indexes arrays stacks queues arrays `Lists`, non repeatable items index values datasets collections `Sets`, scoring weight indexing sorted datasets arrays sequences `Sorted Sets`).
- **Persistence Snapshots Data Recovery Disk Engines:** Structural backup disk dumping files tracking indices data file systems snapshot options engines parameters (jaise RDB point-in-time snapshots, AOF append-only log file streams write paths updates models profiles targets variables maps data protection operations data recovery configurations layers crash proof system settings updates).

21. AWS VPC (VIRTUAL PRIVATE CLOUD)

AWS VPC aapke AWS cloud resources ke liye ek logically isolated virtual network infrastructure setup create karta hai jo aapki computing environments ko absolute isolation aur networking privacy control settings deta hai.

Core Sub-Components Details:

- **VPC Network Boundary IP Block Layout Configuration:** Master networking range definition configurations parameters CIDR blocks frames tracking, jahan total cloud private virtual data centers allocations addresses space configurations parameters options network maps blocks.
- **Network Sub-Divided Routing Segments (`Subnets`):** Granular internal network blocks ranges frames allocations metrics tracking profiles, jahan public accessible targets paths elements maps (`Public Subnets` connected directly internet routing configurations maps gateways structures profiles records fields) aur completely secure isolated backend layers structures blocks resource hosts (`Private Subnets` data layers environments models platforms systems).
- **Network Data Routing Guidance Tables (`Route Tables`):** Logic data routing guidelines criteria directional lines rules structures records files profiles, jahan network traffic flows connections pathways target next-hops targets maps are explicitly mapped definitions profiles.
- **Internet Public Communication Entry Gateway (`Internet Gateway - IGW`):** VPC network edge perimeter communication router gateway engine solutions components blocks, jahan private data centers fields interfaces reach public open external online target spaces channels data flows communication links.
- **Subnet Level Stateless Protection Boundaries (`Network ACLs - NACLs`):** Granular traffic filtering check structures options lists tracks parameters profiles borders, jahan inbound outbound processing streams lines data traffic parameters are checked packets evaluations rules criteria stateless firewall frameworks levels checks configurations.
- **Compute Instances Interfaces Layer Stateful Firewalls (`Security Groups`):** Target specific node level instances compute machines interfaces layer boundaries security controllers packages rules models platforms targets parameters configurations variables lines stateful security firewalls rules controls protections parameters.

- **Private Network Outbound Translation Gateway (`NAT Gateway`):** Specialized network architecture address translators machines proxy nodes engines components layers, jahan private subnet hidden systems computational units fetch out-connections outbound paths open public space links (jaise continuous security dependencies system patches update installation streams code) par external anonymous network target systems nodes tracing attempts blocks connections target elements blocks directly securely completely holds.

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22. AWS EC2 (ELASTIC COMPUTE CLOUD)

AWS EC2 virtual machines compute computing instances scalable infrastructure processing setups provide karne ki ek central platform compute architecture layer service hai.

Technical Components Breakdowns:

- **Pre-Configured System Templates Blueprints (`AMIs - Amazon Machine Images`):** Software image manifests metadata blueprints layouts templates configurations configurations files records, jahan operational platform deployment basic components target operating systems base data files pre-installed software layers system data sets templates configurations maps.
- **Workloads Specific Processing Families Groupings (`Instance Types`):** Computational power specifications scaling classifications metrics tracking profiles families, jahan CPU cores counts, RAM memory size boundaries, network lanes bands parameters, local block storage speeds combinations arrays options families.
- **Persistent High Speed Block Storage Units (`EBS Volumes - Elastic Block Store`):** Local host level compute disks infrastructure storage networks containers blocks blocks parameters, jahan operational compute engine data operations system blocks files reads writes cycles fully high speed local block storage allocations properties data structures profiles records fields settings data.
- **Secure Terminal Access Credentials Codes (`Key Pairs`):** Cryptographic cryptographic key components validation checks credentials profiles parameters security pairs, jahan user administrative terminal logins options (SSH protocol encryption layers mappings) remote servers terminals interactions checks secure authorization rights completely validate controls parameters updates.
- **Static Persistent Network Route Values (`Elastic IPs`):** Permanent public IPv4 routing numbers values parameters records structures maps, jahan cloud setups updates changes variations lifecycle infrastructure transitions dependencies layers instances restarts cycles parameters data completely intact unchange profiles records data formats tracking network lines routing maps parameters.

23. AWS LAMBDA SERVERLESS COMPUTE

AWS Lambda ek physical serverless infrastructure computing architecture engine distribution system platform layer hai jo background instances compute servers provisioning tasks ke bina code automatically execute aur scale karta hai.

Core Components Aur Execution Parameters:

- **Target Script Function Execution Codebase:** Code files syntax algorithms configuration blocks data paths, jahan custom application scripts execution timeout settings computational memory size limits parameters are explicitly mapped definitions profiles.
- **Dynamic Infrastructure Triggers Selectors Configurations:** Event emitters mappings tracking parameters modules indicators hooks options, jahan external platform system actions events changes outputs records (jaise new file data landing indicators streams blocks inside Amazon S3 buckets) active code layers execution loops instant automated invoke triggers checks traces land parameter values check.
- **IAM System Execution Authorizations (`Execution Role`):** Temporary security tokens identities authorization paths permission sets profiles boundaries, jahan running script code models services components secure internal cloud environments resources APIs access keys actions profiles validations framework securely allow permissions execute tasks.
- **Isolated Computing Space (`Execution Environment`):** Low layer execution mini container space environment system runtime instance platform blocks modules tools, jahan code processes dependencies evaluations scripts models platforms run parameters definitions inside safe boundaries computing execution context layers rules blocks units.
- **Reusable Layer Extension Packages Dependencies (`Lambda Layers`):** Modular functional code extensions profiles packages models tools dependencies libraries layer files, jahan shareable libraries code elements configurations dependencies packages options remain isolated apart from main script deploy codes package parameters size optimize steps.
- **Asynchronous Execution Output Routing Targets (`Destinations`):** Result response routing configurations parameters targets metrics tools indicators profiles targets, jahan automated secondary notifications logic tracks run context success codes execution maps ya execution runtime error drops failure handles parameters alerts triggers.

24. AWS IAM (IDENTITY AND ACCESS MANAGEMENT)

AWS IAM poore cloud landscape resources actions permissions ko absolute secure limits control settings ke sath control karne ki framework web service hai.

Logical Structural Elements:

- **Individual Account Identities Parameters (`Users`):** Specific people or distinct custom machine connections identities handlers parameters profiles profiles, jahan target login access strings tracking indicators controls platform access properties configurations variables lines settings.
- **Common Permissions Logical Groups Containers (`Groups`):** Permissions management optimization tracking listings files profiles jahan multiple similar system users credentials mapping groupings structures arrays configurations data profiles options layers.
- **Temporary Platform Identity Assume Rights Mappings (`Roles`):** Identity assumptions profiles specifications parameters security profiles pairs, jahan short lifecycle operational validation rights permissions maps (AWS cloud resources computing engines processes tasks layers mappings) dynamic entities targets rights shifting completely check safe authority parameters.
- **Granular JSON Data Access Permissions Documents (`Policies`):** Programmatic code scripts parameters expressions maps statements data access boundaries files code base layouts profiles schemas blocks inside account governance files (JSON structured syntax systems rules formats files options tracks).

25. AWS RDS (RELATIONAL DATABASE SERVICE)

AWS RDS cloud relational engines configurations instances setups maintain karne ki ek managed database computing database platform system service hai.

Core Database Framework Specs:

- **Database Compute Server Environment Box (`DB Instance`):** Dedicated operational virtual machine engine space computing host database layer, jahan relational operations execution workflows processes memory allocations systems are fully alive processing parameters.
- **Engine Processing Framework Applications Software (`DB Engines`):** Software platform configuration packages database systems parameters engine options, jahan structural SQL logic processing variants rules configurations metrics data (jaise MySQL standard modules, PostgreSQL robust database engines frameworks lines, Oracle corporate setups profiles modules) live active configurations databases operations systems configurations models.
- **Granular Database Engine Settings Properties (`Parameter Groups`):** Configuration variables blueprints parameter parameters profiles templates data records sheets, jahan structural database adjustments fields variables values overrides parameters are explicitly structured track settings properties database optimization profiles parameters.
- **Network Isolation Proxy Firewalls (`Security Groups`):** Database connectivity access permissions layout lists tracking parameters firewalls borders, jahan database ports entries data traffic connections requests are securely checked instances interfaces filtering configurations.
- **Continuous Automated Storage Safety Snapshot Runs (`Automated Backups`):** Completely automated platform state saving routines infrastructure snapshots recovery engines variables (point-in-time database data streams transactional logs capture paths profiles layers data protection parameters safe recovery configurations layers crash proof storage settings).

26. AWS ELB (ELASTIC LOAD BALANCING) SUBSYSTEMS

AWS ELB automatic processing traffic application components traffic inputs distribution layers automate karne ka cloud framework tier hai jo infrastructure processing operations highly resilient and scalable format architectures par distribute karta hai.

Structural Distribution Sub-Units:

- **Incoming Traffic Connection Monitoring Filters (`Listeners`):** Front-end traffic target connectivity entry port check monitors filters configurations parameters, jahan client connection parameters incoming web streams levels definitions triggers run protocols checking routing profiles criteria metrics check.
- **Traffic Routing Condition Rules Configuration Script (`Rules`):** Logical evaluation steps parameters conditions rules manifests scripts data paths, jahan host path parameters, header values matches rules criteria context mapping blocks parameters configurations check metrics values check.
- **Compute Resources Targets Destination Pools Mappings (`Target Groups`):** Logical computational backend targets group configurations layout listings metrics profiles, jahan virtual machines EC2 compute blocks nodes instances targets, microservices containers allocations endpoints pools mappings fully mapped definitions profiles status monitoring diagnostics protections.

Elastic Load Balancer ke Variations:

- **Application Load Balancer (ALB):** OSI model ke Layer 7 (Application Layer) par operate karta hai. Yeh HTTP aur HTTPS traffic ko bohot bariki se samajhta hai aur requests ke advanced parameters (jaise HTTP headers, target request URL paths) ke basis par intelligent load sharing and smart microservices routing profiles ko direct compute groups me balance karta hai.
- **Network Load Balancer (NLB):** OSI model ke Layer 4 (Transport Layer) par kaam karta hai. Yeh extreme high performance, ultra-low latency, aur millions of requests per second ko handle karne ke liye optimize hota hai. Yeh direct raw TCP, UDP, aur TLS network traffic streams ko directly operational worker pools par target balance karta hai.

- **Classic Load Balancer (CLB):** AWS ka legacy load distribution model solutions framework hai jo Layer 4 aur Layer 7 dono par baseline routing options apply karta hai. Yeh simple, standard computing instances ko connect karke fundamental load distribution and basic infrastructure health status alerts setup chalta hai.

27. AWS ROUTE 53 CLOUD DNS

AWS Route 53 highly resilient global scaling internet cloud domain mapping routing protocol suite infrastructure management tracking network layer web DNS service application hai.

Advanced Network Mapping Attributes:

- **Domain Configuration Tracking Boundaries (`Hosted Zones`):** Central primary logical domain management metadata boundaries tracking containers options, jahan public internet domain name entries system zone files records layouts maps configurations parameters options network maps blocks.
- **Target Path Specification Entries Data Records (`Resource Record Sets`):** DNS data paths record specifications configurations maps records syntax templates, jahan alphanumeric internet domain request inputs target location raw cloud parameters numbers IP addresses space values mappings are explicitly mapped definitions profiles data formats.
- **Target Availability Continuous Checking Automation (`Health Checks`):** Continuous endpoints visibility target status check monitoring components blocks variables, jahan real time applications servers computational nodes health check verification paths loops checks automated failures status tracking routes updates models profiles markers triggers.
- **Traffic Engineering Routing Design Policies Options:** Multi target routing logic parameters design blueprints frameworks guidelines data charts paths (Simple, Failover, Geolocation, Latency, Weighted, Multivalue Answer).